

NYAYAVault — Detailed Project Explanation & Page-by-Page Screen Guide

SIH Problem Statement: SIH26190 — Secure Digital Document Management System for Legal and Investigation Documents

Sponsoring Agency: National Crime Records Bureau (NCRB) / Ministry of Home Affairs

Prototype Version: Prototype 1.0

Executive Summary

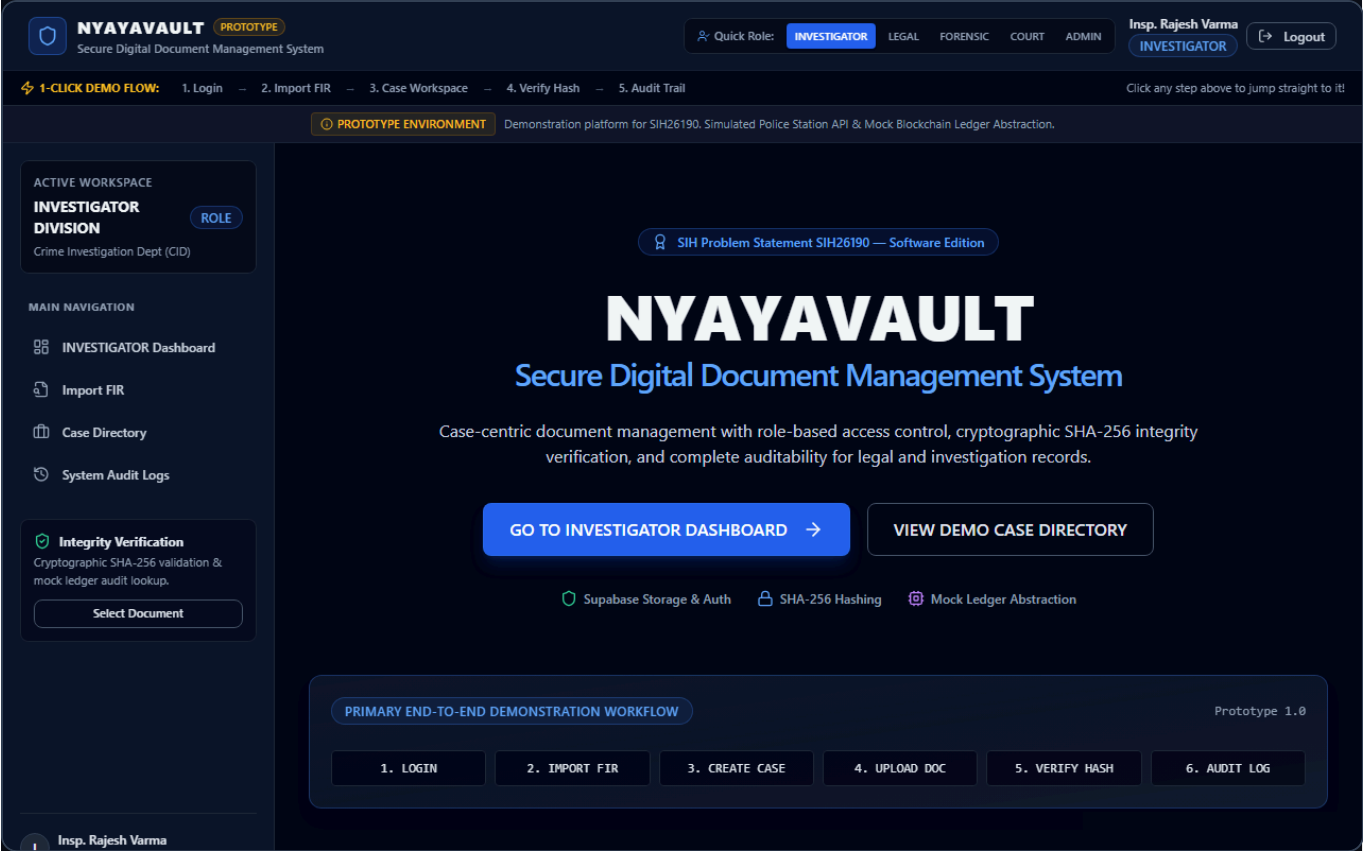
NYAYAVault is a case-centric, role-governed digital document management platform engineered for law enforcement agencies, prosecution departments, forensic laboratories, and judicial courts. It addresses critical challenges in legal document management including document tampering, fragmented evidence, lack of auditability, and unauthorized access.

Core Technical Pillars:

- Case-Centric Document Registry:** All FIRs, witness statements, forensic reports, and charge sheets are indexed under unified investigation case files.
- Cryptographic SHA-256 Fingerprinting:** Calculates a 256-bit hash live in the browser using the Web Crypto API (`crypto.subtle`) before file upload.
- Mock Blockchain Ledger Abstraction:** `BlockchainService` registers deterministic proof transactions (`TX-2026-XXXXXX`) in PostgreSQL `integrity_records` .
- Role-Based Access Control (RBAC):** Enforces access separation across 5 distinct role divisions: `INVESTIGATOR` , `LEGAL` , `FORENSIC` , `COURT` , and `ADMIN` .
- Simulated Police Station API Integration:** Demonstrates external FIR data retrieval (`FIR/2026/01842`) yielding automated case creation.
- Immutable Audit Trail:** Centralized security logging (`AuditService`) capturing user IDs, timestamps, IP addresses, and metadata payloads.

Page-by-Page Detailed Breakdown & Screenshots

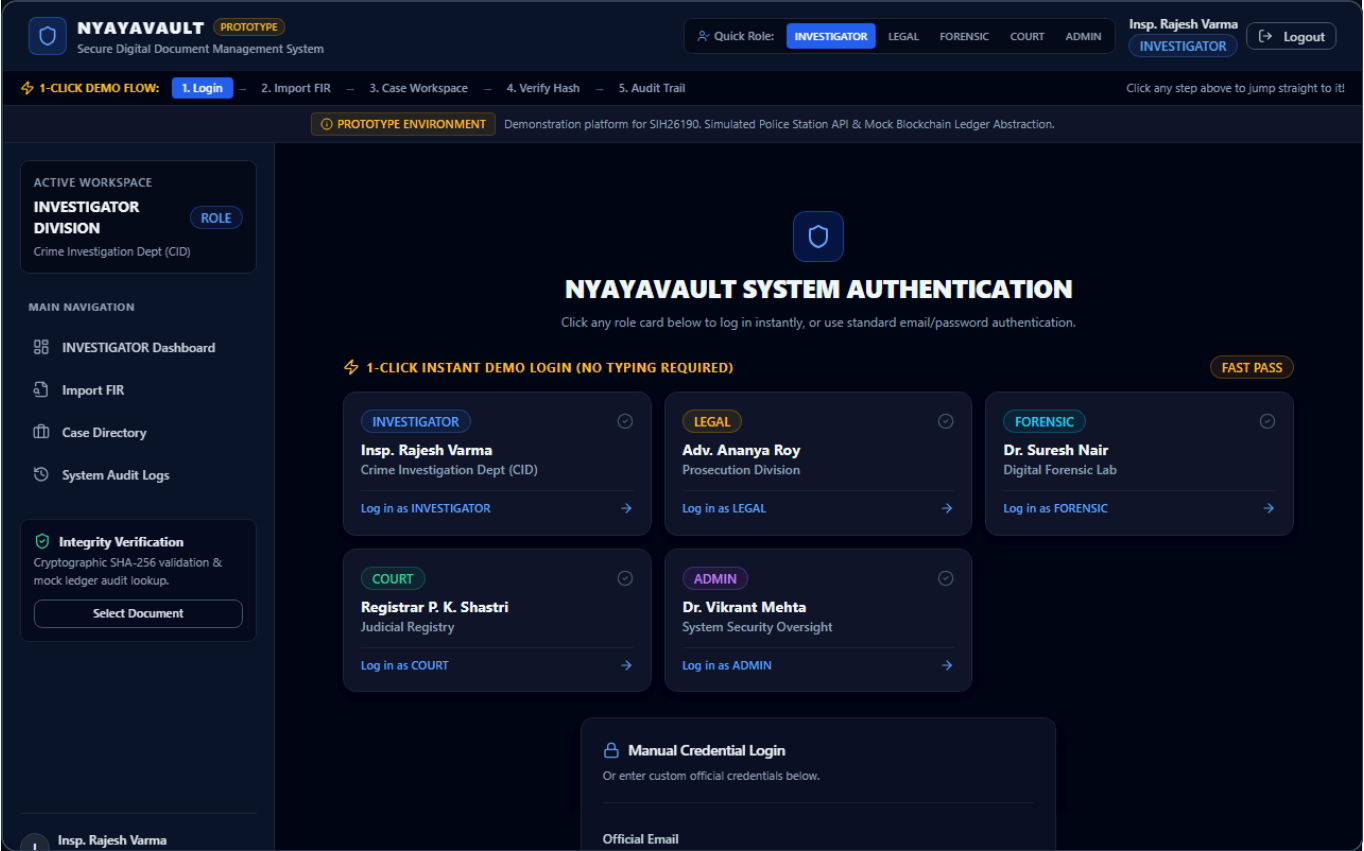
1. Landing Page (/)



Purpose & Capabilities:

- Introduces NYAYAVULT brand, SIH Problem Statement SIH26190 banner, and sponsoring organization (NCRB / Ministry of Home Affairs).
- Displays primary architectural pillars: Secure Access Control, Document Integrity, Case Management, and Immutable Audit Trail.
- Highlights the **Primary 6-Step End-to-End Workflow** (**LOGIN** $\rightarrow$ **IMPORT FIR** $\rightarrow$ **CREATE CASE** $\rightarrow$ **UPLOAD DOC** $\rightarrow$ **VERIFY HASH** $\rightarrow$ **AUDIT LOG**).
- Provides 1-click launch buttons for all 5 role workspaces.

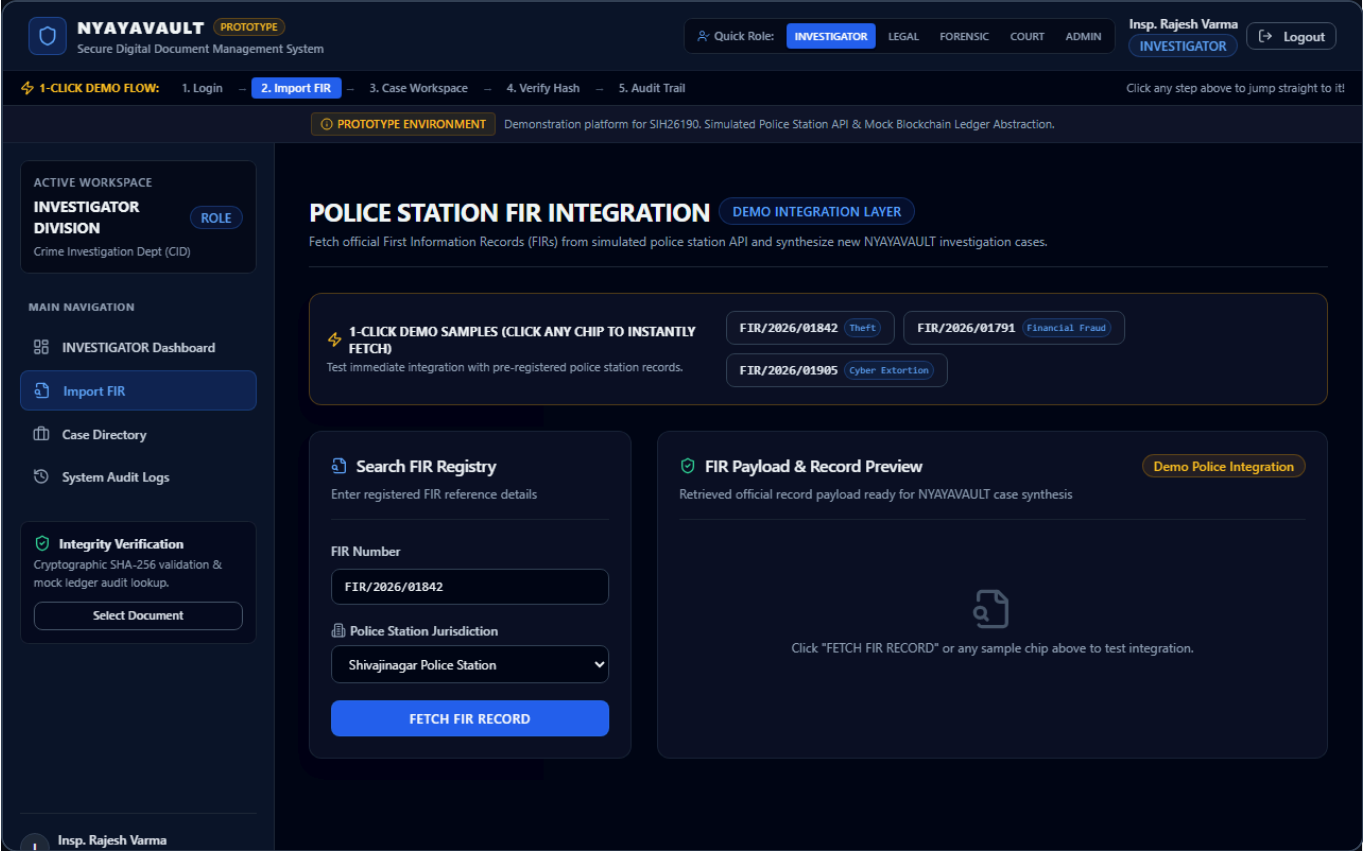
2. Login Portal & 1-Click Role Switcher (/login)



Purpose & Capabilities:

- Authenticates users via Supabase Auth and manages session persistence.
- Features **1-Click Instant Demo Login Cards** for extreme simplicity—clicking any role card (**Insp. Rajesh Varma** , **Adv. Ananya Roy** , **Dr. Suresh Nair** , **Registrar P. K. Shastri** , **Dr. Vikrant Mehta**) logs in immediately without typing.
- Includes standard manual email/password fallback login form.

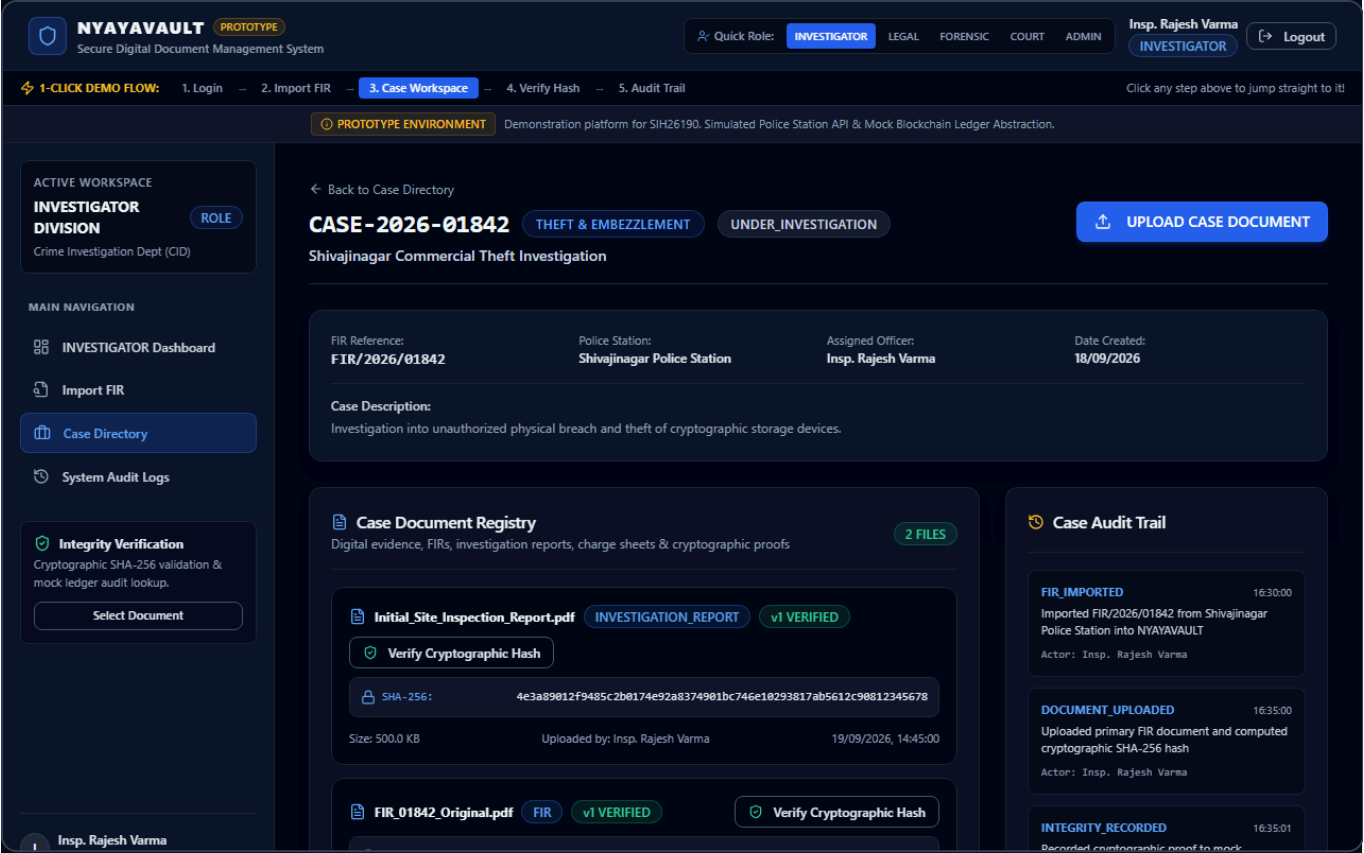
4. Police Station FIR Import Integration (/investigator/fir)



Purpose & Capabilities:

- Demonstrates integration layer fetching First Information Reports from police station databases.
- Includes **1-Click Sample Chips** (`FIR/2026/01842` , `FIR/2026/01791` , `FIR/2026/01905`) that automatically populate and fetch records in 1 tap.
- Multi-step loading state (*"Connecting to Police Station Integration Layer..."* $\rightarrow$ *"FIR Retrieved Successfully"*).
- **"IMPORT INTO NYAYAVULT"** button automatically creates a database case record, writes the FIR record, logs security telemetry, and redirects to the new case workspace.

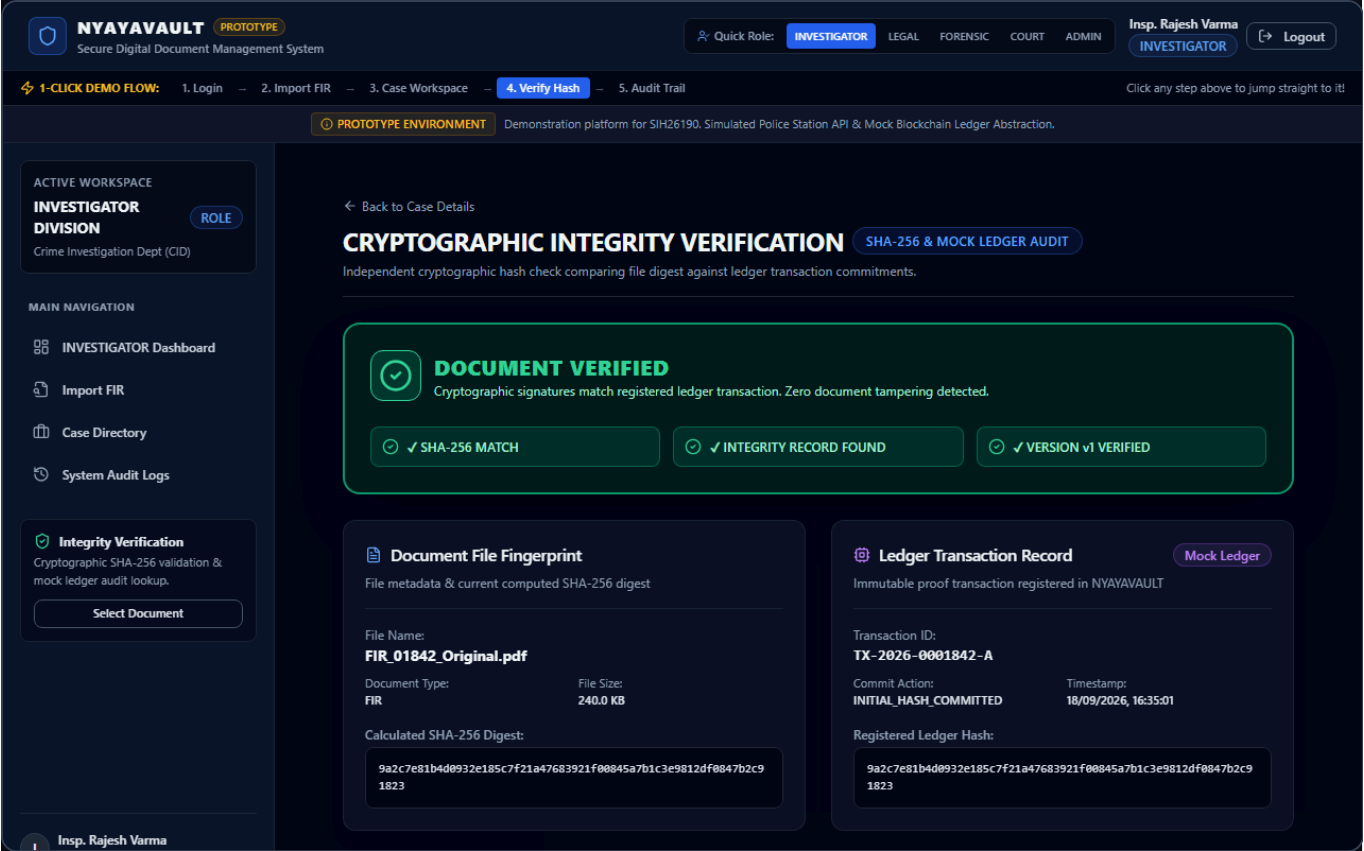
5. Central Case Workspace (/cases/[id])



Purpose & Capabilities:

- Synthesizes all case-level metadata: Case Number, FIR Reference, Police Station Jurisdiction, Assigned Officer, and Creation Date.
- **Case Document Registry:** Categorizes uploaded FIRs, investigation reports, forensic analysis, charge sheets, and evidence records with live SHA-256 hash badges.
- **Document Upload Modal:** Features a  **Add Sample PDF** button that auto-generates a sample PDF file in 1 click, computes its SHA-256 hash live in the browser, uploads to Supabase Storage, and commits proof to the mock ledger.
- **Case Audit Sidebar:** Displays chronological security events tied specifically to this investigation.

6. Cryptographic Document Integrity Verification (/documents/[id]/verify)



Purpose & Capabilities:

- Provides independent cryptographic verification for any stored document.
- Calculates current file hash via Web Crypto API and compares against registered `documents.sha256` and mock ledger transaction ID.
- **SUCCESS UI:** Prominent green banner (`DOCUMENT VERIFIED – ✓ SHA-256 MATCH` , `✓ INTEGRITY RECORD FOUND` , `✓ VERSION VERIFIED`).
- **Tamper Simulation Mode:** Interactive button "Simulate 1-Bit File Alteration" alters 1 character of the hash digest to demonstrate instant red "DOCUMENT INTEGRITY FAILED" alert.

7. System Audit Trail (/audit)



NYAYAVULTPROTOTYPE

Secure Digital Document Management System

Quick Role:

INVESTIGATOR

LEGAL

FORENSIC

COURT

ADMIN

Insp. Rajesh Varma

INVESTIGATOR

Logout

1-CLICK DEMO FLOW:

1. Login

2. Import FIR

3. Case Workspace

4. Verify Hash

5. Audit Trail

Click any step above to jump straight to it!

PROTOTYPE ENVIRONMENT

Demonstration platform for SIH26190. Simulated Police Station API & Mock Blockchain Ledger Abstraction.

ACTIVE WORKSPACE

INVESTIGATOR

DIVISION

Crime Investigation Dept (CID)

ROLE

MAIN NAVIGATION

INVESTIGATOR Dashboard

Import FIR

Case Directory

System Audit Logs

Integrity Verification

Cryptographic SHA-256 validation & mock ledger audit lookup.

Select Document

Insp. Rajesh Varma

SYSTEM AUDIT TRAIL

3 AUDIT RECORDS

Immutable system event log capturing user authentication, FIR imports, document SHA-256 hash commitments, and verification events.

Search audit descriptions, users, cases...

Filter Action: ALL

Security Event Stream

Chronological system activity ordered by timestamp

FIR_IMPORTED

Insp. Rajesh Varma

CASE-2026-01842

18/09/2026, 16:30:00

Imported FIR/2026/01842 from Shivajinagar Police Station into NYAYAVULT

{ "source": "Demo Police Station Integration", "fir_number": "FIR/2026/01842" }

DOCUMENT_UPLOADED

Insp. Rajesh Varma

CASE-2026-01842

18/09/2026, 16:35:00

Uploaded primary FIR document and computed cryptographic SHA-256 hash

{ "sha256": "9a2c7e81b4d0932e185c7f21a47683921f00845a7b1c3e9812df0847b2c91823", "size": 245800 }

INTEGRITY_RECORDED

Insp. Rajesh Varma

CASE-2026-01842

18/09/2026, 16:35:01

Recorded cryptographic proof to mock blockchain ledger with Transaction ID TX-2026-0001842-A

Purpose & Capabilities:

- Centralized security event stream recording all system actions.
- Filters:** Filterable by Action (LOGIN , LOGOUT , FIR_FETCHED , FIR_IMPORTED , CASE_CREATED , DOCUMENT_UPLOADED , HASH_GENERATED , INTEGRITY_RECORDED , DOCUMENT_VERIFIED), User ID, Case ID, and Search Query.
- Displays timestamp, actor profile, target case/doc, action badge, description, and expandable JSON metadata payload.

Page 8 of 13 — Confidential Legal Technology Prototype

8. Legal Prosecution Workspace (/legal)

NYAYAVULT

Secure Digital Document Management System

PROTOTYPE

Quick Role:

INVESTIGATOR

LEGAL

FORENSIC

COURT

ADMIN

Adv. Ananya Roy

LEGAL

Logout

1-CLICK DEMO FLOW:

1. Login

2. Import FIR

3. Case Workspace

4. Verify Hash

5. Audit Trail

PROTOTYPE ENVIRONMENT

Demonstration platform for SIH26190. Simulated Police Station API & Mock Blockchain Ledger Abstraction.

ACTIVE WORKSPACE

LEGAL DIVISION

ROLE

Directorate of Public Prosecutions

MAIN NAVIGATION

LEGAL Dashboard

Case Directory

System Audit Logs

Integrity Verification

Cryptographic SHA-256 validation & mock ledger audit lookup.

Select Document

Adv. Ananya Roy

LEGAL PROSECUTION WORKSPACE

LEGAL OFFICERS & PROSECUTORS

Access authorized investigation cases, review charge sheets and witness statements, and verify document cryptographic integrity.

Assigned Legal Cases

2 Cases

Authorized for Legal Review

Documents to Review

2 Files

Charge Sheets & FIR Filings

Verified Documents

2 Verified

Cryptographically Validated

Authorized Legal Cases

Investigation cases assigned for legal prosecution review

CASE-2026-01842

Theft & Embezzlement

UNDER_INVESTIGATION

Shivajinagar Commercial Theft Investigation

Station: Shivajinagar Police Station | FIR: FIR/2026/01842

Inspect Documents

CASE-2026-01791

Financial Fraud

IN_REVIEW

Pune Division Corporate Fund Embezzlement

Station: Pune Division Central PS | FIR: FIR/2026/01791

Inspect Documents

Purpose & Capabilities:

- Dedicated workspace for Public Prosecutors and Legal Officers.
- Displays authorized investigation cases assigned for legal prosecution review.
- Allows legal teams to inspect charge sheets, witness statements, and verify cryptographic evidence integrity prior to court submission.

Page 9 of 13 — Confidential Legal Technology Prototype

9. State Digital Forensic Laboratory Workspace (/forensic)

NYAYAVULT

Secure Digital Document Management System

PROTOTYPE

Quick Role:

INVESTIGATOR

LEGAL

FORENSIC

COURT

ADMIN

Dr. Suresh Nair

FORENSIC

Logout

1-CLICK DEMO FLOW:

1. Login

2. Import FIR

3. Case Workspace

4. Verify Hash

5. Audit Trail

Click any step above to jump straight to it!

PROTOTYPE ENVIRONMENT

Demonstration platform for SIH26190. Simulated Police Station API & Mock Blockchain Ledger Abstraction.

ACTIVE WORKSPACE

FORENSIC DIVISION

State Digital Forensic Laboratory

MAIN NAVIGATION

FORENSIC Dashboard

Case Directory

System Audit Logs

Integrity Verification

Cryptographic SHA-256 validation & mock ledger audit lookup.

Select Document

Dr. Suresh Nair

STATE DIGITAL FORENSIC LABORATORY

FORENSIC DIVISION

Digital evidence intake, SHA-256 fingerprint logging, forensic report generation, and chain-of-custody verification.

Digital Evidence Received

2 Artifacts

Encrypted & Hashed

Pending Examination

1 Case Drive

In Queue for Analysis

Reports Submitted

4 Reports

Ledger Locked & Signed

Digital Evidence & Artifact Fingerprints

Cryptographic evidence log tied to active investigation cases

FIR_01842_Original.pdf

SHA-256: 9a2c7e81b4d0932e185c7f21a47683921f08845a7b1c3e9812df0847b2c91823

Verify Fingerprint

Initial_Site_Inspection_Report.pdf

SHA-256: 4e3a89012f9485c2b0174e92a8374901bc746e10293817ab5612c90812345678

Verify Fingerprint

Purpose & Capabilities:

- Dedicated workspace for Forensic Examiners and Lab Analysts.
- **Digital Evidence Intake:** Displays digital artifact fingerprints, encrypted drive dumps, and forensic analysis reports.
- Maintains chain-of-custody logging and cryptographic fingerprint verification.

10. District & Sessions Judicial Court Workspace (/court)

NYAYAVULT

Secure Digital Document Management System

PROTOTYPE

Quick Role:

INVESTIGATOR

LEGAL

FORENSIC

COURT

ADMIN

Hon. Registrar P. K. Shastri

COURT

Logout

1-CLICK DEMO FLOW:

1. Login

2. Import FIR

3. Case Workspace

4. Verify Hash

5. Audit Trail

PROTOTYPE ENVIRONMENT

Demonstration platform for SIH26190. Simulated Police Station API & Mock Blockchain Ledger Abstraction.

ACTIVE WORKSPACE

COURT DIVISION

ROLE

District & Sessions Judiciary

MAIN NAVIGATION

COURT Dashboard

Case Directory

System Audit Logs

Integrity Verification

Cryptographic SHA-256 validation & mock ledger audit lookup.

Select Document

DISTRICT & SESSIONS JUDICIARY REGISTRY

COURT JUDICIARY WORKSPACE

Judicial registry for reviewing verified charge sheets, FIR filings, and validating cryptographic document authenticity before court hearings.

Cases Received for Hearing

2 Cases

Judicial Registry Synced

Pending Judicial Verification

0 Pending

All Filings Hash Verified

Finalized Cases

1 Case

Adjudication Complete

Judicial Case Filings

Verified case records ready for judicial bench examination

CASE-2026-01842

COURT DOCKET

Shivajinagar Commercial Theft Investigation

Police Station: Shivajinagar Police Station | FIR: FIR/2026/01842

Review Court Docket

CASE-2026-01791

COURT DOCKET

Pune Division Corporate Fund Embezzlement

Police Station: Pune Division Central PS | FIR: FIR/2026/01791

Review Court Docket

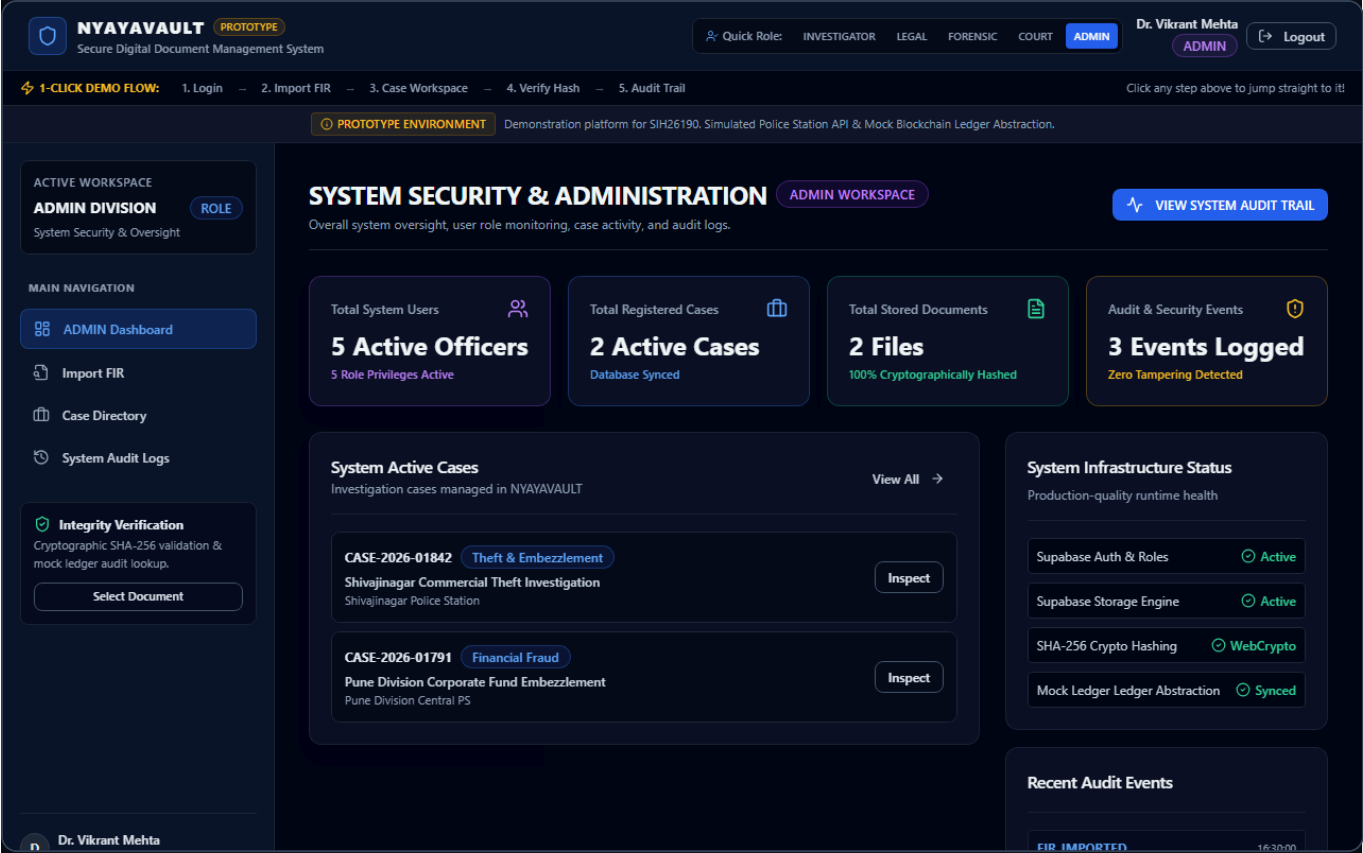
Hon. Registrar P. K. Shastri

Purpose & Capabilities:

- Dedicated workspace for Judges, Judicial Registrars, and Bench Clerks.
- Displays court dockets, verified charge sheets, and trial hearing schedules.
- Ensures all submitted evidence records are cryptographically validated prior to judicial review.

Page 11 of 13 — Confidential Legal Technology Prototype

11. System Security & Administration Workspace (/admin)



Purpose & Capabilities:

- Central command dashboard for System Security Administrators.
- **Metrics Grid:** Total System Users, Registered Cases, Stored Documents, and Security Audit Events.
- **Infrastructure Status:** Health indicators for Supabase Auth, Storage Engine, WebCrypto Hashing, and Mock Ledger Sync.

⚡ 1-Click Simplicity Features Summary

NYAYAVULT is designed for instant evaluation with zero friction:

1. **Header Workflow Stepper:** Click any step ([1. Login](#) → [2. Import FIR](#) → [3. Case Workspace](#) → [4. Verify Hash](#) → [5. Audit Trail](#)) in the top navbar to jump straight to it.
2. **1-Click Instant Login:** Log in as any role with a single click.
3. **1-Click Sample FIR Chips:** Fetch FIR records instantly with 1 tap.
4. **1-Click Sample PDF Generator:** Create evidence PDF files on the fly without searching your local hard drive.
5. **1-Click Tamper Simulation:** Test tamper detection with 1 button toggle.

Running the Project Locally

```
# Install dependencies
npm install

# Start development server
npm run dev
```

Open `http://localhost:3000` in your browser.